Real-Time Time-Dependent Density Functional Theory for Periodic and Spin-Polarized Systems with the Real-Space Multigrids (RMG)[†]

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Abstract

[Abstract: to be written last, after all sections are complete. Approximately 200 words. Should mirror paper 1 abstract structure: (1) what we present, (2) what problem it solves, (3) key methodological contribution, (4) validation/benchmarking, (5) scalability demonstration and outlook.]

1 Introduction

Real-time time-dependent density functional theory (RT-TDDFT) has established itself as one of the primary tools for simulating the electronic response of materials to external electromagnetic perturbations.[1,2] Its applications span a broad range of phenomena, including optical spectroscopy,[3–7] charge transfer and photovoltaics,[2,8,9] nanoplasmonics,[10–12] and spin and magnetization dynamics.[13–15] In an earlier publication, we presented a scalable RT-TDDFT implementation within the Real-space MultiGrid (`RMG`) code for finite molecular systems, demonstrating simulations of up to 24,000 electrons on massively parallel architectures.[?] The present work extends that implementation to periodic systems and spin-polarized calculations, significantly broadening the range of physically relevant problems accessible to `RMG`.

The transition from finite to periodic systems in RT-TDDFT requires a corresponding transition in the treatment of the electromagnetic perturbation. In finite systems, the length gauge—in which the coupling to an electric field is expressed through the dipole operator $\hat{\mathbf{r}}$—is natural and convenient. In periodic systems, however, the position operator is ill-defined because it is not translationally invariant, and the dipole moment of an extended system is intrinsically multivalued.[?,?] The correct framework for periodic systems is the velocity gauge, in which the electromagnetic interaction is expressed through the vector potential $\mathbf{A}(t)$ and the current operator, which is inherently translationally invariant. The transition between the two gauges is effected by a gauge transformation of the time-dependent Kohn-Sham equations,[?,?,?] and requires careful treatment of the nonlocal part of pseudopotentials.[?,?,?] The primary observable in the velocity gauge is the macroscopic current density $\mathbf{J}(t)$, from which the optical conductivity $\boldsymbol{\sigma}(\omega)$ and dielectric function $\boldsymbol{\varepsilon}(\omega)$ are extracted by Fourier analysis.

RT-TDDFT for periodic systems has been implemented in several codes. The `Octopus` code,[16] based on real-space grids, provides a comprehensive periodic RT-TDDFT implementation including both length and velocity gauges and has served as the primary validation reference for the present work. Plane-wave codes including `Qbox`/`Qball`,[17,18] `cp2k`,[19] and `Exciting`[?] offer periodic RT-TDDFT with natural handling of Born-von Kármán boundary conditions. Atom-centered basis set codes such as `SIESTA`[20] and `DFTB+`[21] provide efficient implementations for systems where localized orbitals are appropriate. The `DFT-FE` code[22] offers an alternative real-space approach based on finite elements. [Check: add any missing periodic RT-TDDFT codes before submission. In particular: GPAW, Abinit, Yambo.]

The treatment of spin-polarized systems within RT-TDDFT introduces an additional degree of freedom: the electronic density is resolved into spin-up and spin-down components, and the time-dependent exchange-correlation potential depends on both. In the collinear approximation, the two spin channels evolve independently subject to a common self-consistent spin-dependent potential, and the optical response carries spin-specific signatures. This capability is essential for magnetic insulators such as CrI_3, where the spin-up and spin-down optical spectra differ markedly and encode information about the electronic and magnetic structure.[13,15] The non-collinear spin extension, in which the electronic states are represented as two-component spinors, is beyond the scope of the present work and will be addressed in a future publication.

A distinctive feature of the `RMG` implementation is its scalability. The real-space multigrid approach assigns different regions of space to different processors, enabling near-linear scaling to hundreds or thousands of GPU-accelerated nodes.[23,?] This scalability is particularly significant for periodic calculations with spin, where both the k-point sampling and the spin-polarized density matrix increase the computational cost relative to finite systems. To demonstrate this capability, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond—a prototypical point defect system of substantial interest for quantum sensing and quantum information applications.[?] By systematically increasing the supercell size from [X] to [Y] atoms, we show how the optical response of the defect converges as the interaction between periodic images becomes negligible, identifying the minimum supercell size required for physically meaningful results at realistic defect concentrations.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework, beginning with the interaction of matter with electromagnetic radiation in the long-wavelength approximation, proceeding through the gauge transformation from length to velocity gauge, the definition of the current operator including nonlocal pseudopotential contributions, the density matrix propagation scheme, the spin-collinear extension, and the postprocessing route from current to optical conductivity. Section 3 describes the algorithm as implemented in `RMG` and identifies the key approximations and their physical justification. Section 4 presents benchmarking calculations for one-dimensional (H_2 chain), two-dimensional (hexagonal BN), and three-dimensional (Si) periodic systems, followed by spin-polarized calculations for CrI_3, with comparisons to `Octopus` in all cases. The section concludes with the NV-center in diamond scalability demonstration. The paper concludes with a Summary and Outlook.

2 Methodology

All equations in this section are given in atomic units ($\hbar = m_e = e = 4\pi\varepsilon_0 = 1$) unless stated otherwise.

2.1 Overview of RMG

[Brief overview of RMG: real-space grid, multigrid acceleration, GPU scalability. Refer heavily to paper 1[?] and the RMG code paper[23] rather than repeating. Keep to 1–2 paragraphs. Focus on what is new or relevant for the periodic case: k-point sampling, Bloch states, periodic boundary conditions. Mention norm-conserving pseudopotentials (NCPP only).]

[Note from CONTEXT.md: the code uses norm-conserving PP only (asserted in RmgTddft.cpp line 287). Ultrasoft PP requires S≠1 in the propagation (eldyn_nonort.cpp path) and is not yet implemented. State this as an explicit scope limitation here.]

2.2 Interaction of Matter with Electromagnetic Radiation

[DERIVATION NEEDED from Jacek's LaTeX files. Content outline: - Start from the full minimal-coupling Hamiltonian with EM field treated as a plane wave: $\exp(\imath\omega t - \imath\mathbf{q}\cdot\mathbf{r})$ - Derive the long-wavelength (dipole) approximation: $\mathbf{q} \to 0$, i.e., $\lambda \gg$ system size, valid for optical/UV and unit-cell scales - Show that in this limit the vector potential $\mathbf{A}$ becomes spatially uniform: $\mathbf{A}(\mathbf{r}, t) \to \mathbf{A}(t)$ - Result: the two natural gauge choices in the long-wavelength limit are the length gauge (scalar potential $\Phi = -\mathbf{r}\cdot\mathbf{E}$, $\mathbf{A} = 0$) and the velocity gauge ($\Phi = 0$, $\mathbf{A}(t)$ uniform) - Physical interpretation: length gauge couples to position (dipole), velocity gauge couples to momentum (current) - Reference: Mattiat & Luber (2022) Section 2.3; Yabana & Bertsch (1996)]

The time-dependent Kohn-Sham (TDKS) equations governing the electronic dynamics under an external electromagnetic perturbation take the form

$$\imath\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \hat{H}_{\mathrm{KS}}(t)\,\psi_n(\mathbf{r}, t), \tag{1}$$

where $\hat{H}_{\mathrm{KS}}(t)$ is the time-dependent Kohn-Sham Hamiltonian. [Expand: write out $\hat{H}_{\mathrm{KS}}(t)$ with all terms including the EM coupling, then derive the long-wavelength limit.]

2.3 Gauge Transformation: Length to Velocity Gauge

[DERIVATION NEEDED from Jacek's LaTeX files. Content outline: - State clearly why length gauge fails for periodic systems: the position operator $\hat{\mathbf{r}}$ is not translationally invariant; dipole moment of a periodic system is multivalued (refer to modern theory of polarization: Resta 1994, King-Smith & Vanderbilt 1993) - Present the Göppert-Mayer gauge transformation explicitly: gauge function $\chi(\mathbf{r}, t) = \mathbf{r}\cdot\mathbf{A}(t)$, transformations of potentials and wavefunctions (Mattiat eqs. 15–19) - Show that the length gauge TDKS (eq. 2) transforms to the velocity gauge TDKS (eq. 3) - CRITICAL: the nonlocal pseudopotential V^nl does NOT commute with the gauge transformation. The gauge-transformed V^nl acquires an additional term (Mattiat eq. 23). Derive this explicitly here or refer to Appendix A. - Result: velocity gauge TDKS Hamiltonian with A(t) coupling - Reference: Mattiat & Luber (2022) Sections 2.3.1, Appendix A, B; Pickard & Mauri (2001)]

In the length gauge, the TDKS equations read

$$\imath\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \left[\hat{H}_0 + \mathbf{r}\cdot\mathbf{E}(t)\right]\psi_n(\mathbf{r}, t), \tag{2}$$

where $\hat{H}_0$ is the unperturbed Kohn-Sham Hamiltonian and $\mathbf{E}(t)$ is the electric field. Applying the gauge transformation $\chi(\mathbf{r}, t) = \mathbf{r}\cdot\mathbf{A}(t)$, where the vector potential satisfies $\mathbf{E}(t) = -\partial\mathbf{A}/\partial t$, the velocity gauge TDKS equations take the form

$$\imath\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \left[\frac{1}{2}\left(\hat{\mathbf{p}} + \mathbf{A}(t)\right)^2 + V(\mathbf{r}, t) + \tilde{V}^{\mathrm{nl}}(\mathbf{r}, t)\right]\psi_n(\mathbf{r}, t) \tag{3}$$

where $\tilde{V}^{\mathrm{nl}}$ denotes the gauge-transformed nonlocal pseudopotential.[?,?] [Expand: write out all terms of $\tilde{V}^{\mathrm{nl}}$ explicitly. Derive the linearization in A(t) that gives the perturbative coupling term. Connect to Appendix A.]

2.4 Vector Potential and Magnetic Perturbations

[DERIVATION NEEDED from Jacek's LaTeX files. Content outline: - Show that the velocity gauge framework with $\mathbf{A}(t)$ naturally accommodates both electric and magnetic perturbations - Electric perturbation in the long-wavelength limit: $\mathbf{A}(t)$ spatially uniform, $\mathbf{B} = 0$ - Magnetic perturbation (symmetric gauge): $\mathbf{A}(\mathbf{r}) = \frac{1}{2}\mathbf{B} \times (\mathbf{r} - \mathbf{R}_G)$, spatially dependent, introduces gauge origin $\mathbf{R}_G$ - Note: magnetic perturbation sets up ECD/MCD as future extension (not implemented in present work but theoretically grounded here) - The gauge origin dependence with finite basis sets and the standard remedies (GIAO, IGLO) – brief mention, refer to Mattiat for details - Reference: Mattiat & Luber (2022) Section 2.3.2; Pickard & Mauri (2001)]

[Note: This subsection establishes the theoretical foundation for magnetic perturbations. The current paper implements only electric perturbations (A(t) uniform). ECD/MCD calculations are future work. State this explicitly at the end of the subsection.]

2.5 Current Operator and Nonlocal Pseudopotential Contributions

[DERIVATION NEEDED from Jacek's LaTeX files.] This is the most important new subsection. Full derivation required. Content outline:

(a) Canonical vs mechanical momentum: - Canonical: $\hat{\mathbf{p}} = -\imath\nabla$ - Mechanical: $\hat{\boldsymbol{\pi}} = \hat{\mathbf{p}} + \mathbf{A}(t)$ - Velocity operator: $\hat{\mathbf{v}} = (1/\imath)[\hat{\mathbf{r}}, \hat{H}]$ - When $V^{\mathrm{nl}} \neq 0$: $\hat{\mathbf{v}} \neq \hat{\mathbf{p}}/m$ – extra commutator term $[\hat{\mathbf{r}}, V^{\mathrm{nl}}]$ - Ref: Starace (1971,1973); Mattiat & Luber eq.12

(b) Paramagnetic and diamagnetic current: - Full current: $\hat{\mathbf{J}} = \hat{\mathbf{J}}_{\mathrm{para}} + \hat{\mathbf{J}}_{\mathrm{dia}}$ - Paramagnetic: $\hat{\mathbf{J}}_{\mathrm{para}} \propto \mathrm{Re}[\psi^*(-\imath\nabla)\psi]$ - Diamagnetic: $\hat{\mathbf{J}}_{\mathrm{dia}} \propto \mathbf{A}(t)|\psi|^2$ - J_dia = 0 after t=0 in our perturbative scheme (A=0 for t>0) - Justify this explicitly as valid in linear response regime

(c) Continuity equation: - $\partial\rho/\partial t + \nabla\cdot\mathbf{J} = 0$ - Use as consistency/sanity check for the current operator

(d) Current in density matrix formulation: - Macroscopic current: $J_\alpha(t) = \mathrm{Re}\,\mathrm{Tr}[\mathbf{P}(t)\cdot\mathbf{J}^\alpha]$ - Current operator matrix: $J^\alpha_{ab} = p^\alpha_{ab} + [\hat{r}_\alpha, V^{\mathrm{nl}}]_{ab}$ - where $p^\alpha_{ab} = \langle\psi_a|(-\imath\nabla_\alpha + k_\alpha)|\psi_b\rangle$ (the k_α term comes from the Bloch theorem: see eq.XX) - This is what VecPHmatrix() + CurrentNlpp() compute in the code

(e) Nonlocal PP contribution $[r, V^n l]$: - For Kleinman-Bylander separable pseudopotentials: $V^{\mathrm{nl}} = \sum_{I,lm} |\beta_I^{lm}\rangle h^l_{ij}\langle\beta_I^{lm}|$ - Compute $[\hat{r}_\alpha, V^{\mathrm{nl}}]$ explicitly: $[\hat{r}_\alpha, V^{\mathrm{nl}}]|\psi\rangle = \sum_{I,lm} |\beta_I^{lm}\rangle h^l_{ij}\langle[\hat{r}_\alpha, \beta_I^{lm}]|\psi\rangle$ - The commutator acts on the projector: $[\hat{r}_\alpha, \beta_I^{lm}]$ is computed by AppNls_0xyz() in the code - Extended algebra in Appendix A - Ref: Mattiat & Luber (2022) Section 2.2, eq.12; Pickard & Mauri (2001)

VERIFICATION NEEDED: trace the sign/i-factor from theory to code. VecPHmatrix() stores I_t * block_matrix (I_t = i) in Pxyz. CurrentNlpp() adds I_t * block_matrix. Current extracted as Re(zdotc(P, Px)). The -i from the gradient operator + i from storage = real observable. Verify this chain explicitly in the derivation. See Gap 1 in CONTEXT.md.

The velocity operator for a system described by a Hamiltonian containing a nonlocal pseudopotential V^{nl} is obtained from the Heisenberg equation of motion,

$$\hat{v}_\alpha = \frac{1}{\imath}[\hat{r}_\alpha, \hat{H}] = \hat{p}_\alpha + \frac{1}{\imath}[\hat{r}_\alpha, V^{\mathrm{nl}}], \tag{4}$$

where the second term on the right-hand side arises because $[\hat{r}_\alpha, V^{\mathrm{nl}}] \neq 0$ for a nonlocal operator.[?][?] [Expand into full derivation of current operator, paramagnetic and diamagnetic decomposition, density matrix formulation, and explicit KB pseudopotential commutator. Follow derivation philosophy: complete first, then decide what moves to appendix.]

The macroscopic current density is expressed in terms of the one-particle density matrix $\mathbf{P}(t)$ as

$$J_\alpha(t) = \mathrm{Re}\,\mathrm{Tr}[\mathbf{P}(t)\cdot\mathbf{J}^\alpha]\,, \tag{5}$$

where $\mathbf{J}^\alpha$ is the matrix of current operator elements in the Kohn-Sham orbital basis,

$$J^\alpha_{ab} = \langle\psi_a|\left(-\imath\nabla_\alpha + k_\alpha + \frac{1}{\imath}[\hat{r}_\alpha, V^{\mathrm{nl}}]\right)|\psi_b\rangle. \tag{6}$$

[The k_α term in eq. 6 arises from the Bloch theorem: for a periodic orbital $\psi_{n,\mathbf{k}}(\mathbf{r}) = e^{\imath\mathbf{k}\cdot\mathbf{r}}u_{n,\mathbf{k}}(\mathbf{r})$, the gradient operator acting on the full Bloch function gives $\nabla(e^{\imath \mathbf{k}\cdot\mathbf{r}}u) = e^{\imath \mathbf{k}\cdot\mathbf{r}}(\nabla + \imath\mathbf{k})u$. Derive this step explicitly.]

2.6 Density Matrix Propagation

[Adapt from paper 1 (Jakowski et al. 2025, Section 2.2).] The derivation is identical to paper 1 for the case S=1 (orthogonal basis at Gamma point). For $k \neq Gamma$, H is complex-valued (Hermitian but not real-symmetric) and Omega is also complex. The BCH expansion is the same but with complex matrix arithmetic. Key additions vs paper 1: - Note explicitly that for $k \neq Gamma$ the Hamiltonian matrix $\mathbf{H_k} = \langle\psi_a|\hat{H}|\psi_b\rangle$ is complex Hermitian - The BCH propagator (Ieldyn=1) handles complex matrices; the diagonalization path (Ieldyn=2) does NOT – state this - The self-consistency loop (predictor-corrector) is unchanged - The Hamiltonian at each time step includes the A(t)*p coupling term (for VECTOR_POT mode); in the perturbative kick scheme A appears only in the initial H_0 setup Reference sections from paper 1: Magnus expansion and density matrix propagation subsections. Cite: Jakowski & Morokuma (2009); Jakowski et al. JCTC 2025.

The time evolution of the one-particle density matrix $\mathbf{P}(t)$ is governed by the Liouville-von Neumann equation

$$\frac{\partial \mathbf{P}(t)}{\partial t} = -\imath\left[\mathbf{H}(t),\, \mathbf{P}(t)\right], \tag{7}$$

where $\mathbf{H}(t)$ is the Kohn-Sham Hamiltonian matrix in the basis of ground-state Kohn-Sham orbitals. [Continue with Magnus expansion (refer to paper 1 eqs. and cite), then commutator expansion for P(t+dt), self-consistency loop. For complex H: add note about BCH-only path.]

[Note from code inspection (CONTEXT.md Gap 4): magnus() in the code is documented as real-only, but for k$\neq \Gamma$ the call site casts to double*. This is mathematically correct since the Magnus operation $\Omega = -\imath \cdot \frac{1}{2}(H_0 + H_1)\Delta t$ is element-wise and valid for both real and complex H, but should be stated explicitly in the text.]

2.7 Spin-Polarized Extension

[DERIVATION NEEDED from Jacek's LaTeX files. Content outline: - Collinear spin-polarized DFT: two independent sets of KS orbitals $\{\psi_n^{\uparrow}\}$ and $\{\psi_n^{\downarrow}\}$, each with its own density matrix $\mathbf{P}^{\uparrow}(t)$ and $\mathbf{P}^{\downarrow}(t)$ - Spin-resolved electron density: $\rho^{\uparrow}(\mathbf{r},t)$ and $\rho^{\downarrow}(\mathbf{r},t)$ - Total density: $\rho = \rho^{\uparrow} + \rho^{\downarrow}$ - Spin magnetization: $m_s = \rho^{\uparrow} - \rho^{\downarrow}$ - Coupling between spin channels: through the spin-dependent XC potential $V_{\mathrm{xc}}^{\sigma}[\rho^{\uparrow},\rho^{\downarrow}]$ (LSDA or GGA spin-polarized functional) - Each spin channel propagates independently: $\partial\mathbf{P}^{\sigma}/\partial t = -\imath[\mathbf{H}^{\sigma}(t),\mathbf{P}^{\sigma}(t)]$ - The Hamiltonians $\mathbf{H}^{\uparrow}$ and $\mathbf{H}^{\downarrow}$ differ only through $V_{\mathrm{xc}}^{\uparrow} \neq V_{\mathrm{xc}}^{\downarrow}$ - Spin-resolved current and optical conductivity: $J_{\alpha}^{\sigma}(t) = \mathrm{Re}\,\mathrm{Tr}[\mathbf{P}^{\sigma}(t)\cdot\mathbf{J}^{\alpha}]$ $\boldsymbol{\sigma}^{\sigma}(\omega)$ for each spin channel - Total and spin currents: $\mathbf{J}_{\mathrm{charge}} = \mathbf{J}^{\uparrow} + \mathbf{J}^{\downarrow}$ $\mathbf{J}_{\mathrm{spin}} = \mathbf{J}^{\uparrow} - \mathbf{J}^{\downarrow}$ - Note: non-collinear spin (two-component spinors) is beyond the scope of this work and will be presented in a future publication. - Note from code inspection: spin is handled via separate MPI communicator groups (pct.spinpe); spin channels communicate only when updating V_{xc} (rho.get_oppo()).]

In the collinear spin-polarized extension, the density matrix propagation (Section 2.6) is performed independently for each spin channel $\sigma \in \{\uparrow,\downarrow\}$,

$$\frac{\partial \mathbf{P}^{\sigma}(t)}{\partial t} = -\imath\left[\mathbf{H}^{\sigma}(t),\, \mathbf{P}^{\sigma}(t)\right], \tag{8}$$

where the spin-dependent Hamiltonian $\mathbf{H}^{\sigma}(t)$ contains the spin-resolved exchange-correlation potential $V_{\mathrm{xc}}^{\sigma}[\rho^{\uparrow}(\mathbf{r},t),\rho^{\downarrow}(\mathbf{r},t)]$. [Continue: derive spin-resolved current, define total and spin currents, note that J_spin = J_up - J_down is available as a low-cost observable.]

2.8 Postprocessing: Optical Conductivity and Berry Phase

[Content outline:] (a) From J(t) to optical conductivity: - Fourier transform: $\tilde{J}_{\alpha}(\omega) = \int J_{\alpha}(t)e^{\imath\omega t}dt$ - Optical conductivity tensor: $\sigma_{\alpha\beta}(\omega) = \tilde{J}_{\alpha}(\omega)/E_{\beta}(\omega)$ - Dielectric function: $\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \imath\sigma_{\alpha\beta}(\omega)/\varepsilon_0\omega$ - In practice: apply delta-kick perturbation in direction β, Fourier transform $J_\alpha(t)$, divide by $E_\beta(\omega) = E_0$ (constant for delta kick) - Mention damping/broadening: apply $e^{-\gamma t}$ window before FFT

(b) Berry phase polarization as alternative observable: - For periodic systems, polarization can also be extracted via the King-Smith/Vanderbilt Berry phase formula:[?] $\mathbf{P}_{\text{BP}} = \frac{e}{(2\pi)^3}\int_{\text{BZ}} \mathbf{A}_n(\mathbf{k})\, d\mathbf{k}$ where $\mathbf{A}_n = \imath\langle u_{n\mathbf{k}}|\nabla_{\mathbf{k}}|u_{n\mathbf{k}}\rangle$ is the Berry connection - For the Gamma-point supercell limit, the discrete formula (Resta 1998, eq. 48 in Mattiat & Luber): $\langle \hat{d}_\alpha\rangle = \frac{eL}{2\pi}\mathrm{Im}\ln\det S_\alpha$ where $S_{\alpha,ij} = \langle\psi_i|e^{-2\pi\imath r_\alpha/L}|\psi_j\rangle$ - In density matrix form: $P_{\text{BP},\alpha}(t) = \mathrm{Tr}[\mathbf{P}(t)\cdot\mathbf{X}^\alpha]$ where $\mathbf{X}^\alpha$ is the Berry phase position matrix - Equivalence with current-integration result: $\Delta P_\alpha = \int_0^t J_\alpha(t')\, dt'$ (test of internal consistency) - Implementation in RMG: BP_Xml matrix computed by Rmg_BP->tddft_Xml() (see code, RmgTddft.cpp lines 514,517) - Reference: King-Smith & Vanderbilt (1993); Resta (1998); Mattiat & Luber (2022) Section 2.5

The macroscopic current density $\mathbf{J}(t)$ obtained from eq. (5) provides direct access to the optical response of the periodic system. After applying a momentum kick of amplitude E_0 along the Cartesian direction β at $t = 0$, the optical conductivity tensor is obtained from the Fourier transform of the current density as

$$\sigma_{\alpha\beta}(\omega) = \frac{\tilde{J}_\alpha(\omega)}{E_\beta(\omega)} = \frac{1}{E_0}\int_0^\infty J_\alpha(t)\, e^{\imath\omega t-\gamma t}\, dt, \tag{9}$$

where γ is a broadening parameter.[?] [Expand: relate σ to ε; describe how three independent kick simulations (x, y, z) give the full tensor; note crystal symmetry reduces the number of independent components.]

[Add: Berry phase derivation and density matrix form. Show equivalence with current integration. Note that both observables are computed in the present implementation and their agreement serves as an internal consistency check.]

3 Implementation

The implementation of the periodic and spin-polarized RT-TDDFT presented in this work is built on the existing finite-system RT-TDDFT module of `RMG`,[?] which is described in detail in the accompanying paper. Here we focus on the extensions required for periodic boundary conditions and spin-polarized calculations, the key approximations and their physical justification, and the computational organization of the velocity gauge propagation.

3.1 Algorithm

The computational scheme for periodic RT-TDDFT in `RMG` is summarized in Algorithm 1. It follows the same density matrix propagation framework as the finite-system implementation[?] but with three key modifications: (1) the perturbation is applied as a momentum kick via the vector potential $\mathbf{A}$ rather than an electric field dipole kick; (2) the current operator matrix replaces the dipole matrix as the primary observable; and (3) the Hamiltonian matrix is evaluated at each k-point over the Brillouin zone.

3.2 Key Approximations and Their Justification

The implementation described in Algorithm 1 embodies several approximations with respect to the full theory presented in Section 2. We discuss each in turn.

Perturbative momentum kick. The full velocity gauge Hamiltonian (eq. 3) contains the vector potential $\mathbf{A}(t)$ at every time step, requiring the $\mathbf{A}(t)\cdot\hat{\mathbf{p}}$ term to be re-evaluated throughout the propagation. In the present implementation, we instead apply $\mathbf{A}(t)$ as a delta-function kick at $t = 0$ only: the momentum kick is encoded in the initial Hamiltonian $\mathbf{H}_{-1}$ by adding the $\mathbf{A}(t_0)\cdot\mathbf{J}^\alpha$ matrix (step 6 in Algorithm 1), after which $\mathbf{A} = 0$ for all $t > 0$. This approach is equivalent to first-order perturbation theory in $|\mathbf{A}_0|$, and it provides the full linear optical response of the system through

Algorithm 1 Periodic RT-TDDFT Algorithm (VECTOR_POT mode)

/* **Ground state DFT calculations:** */
Generate Bloch orbitals $|\psi_{n,\mathbf{k}}\rangle$, Kohn-Sham Hamiltonian $\mathbf{H}^{\sigma}_{0,\mathbf{k}}$ for each k-point $\mathbf{k}$ and spin σ
Select active space: N_{occ} occupied + N_{virt} virtual orbitals

/* **Build current operator matrices (once, before time loop):** */
For each k-point $\mathbf{k}$ and spin σ:
  Build paramagnetic current matrix: $J^{\alpha}_{ab} \leftarrow \langle\psi_a|(-\imath\nabla_\alpha + k_\alpha)|\psi_b\rangle$ [VecPHmatrix()]
  Add nonlocal PP correction: $J^{\alpha}_{ab} \leftarrow J^{\alpha}_{ab} + \langle\psi_a|[\hat{r}_\alpha, V^{\text{nl}}]|\psi_b\rangle$ [CurrentNlpp()]
  Add $\mathbf{A}(t_0)\cdot\mathbf{J}$ to $\mathbf{H}_{-1,\mathbf{k}}$ to encode the perturbation kick at $t = 0$

/* **Initialize density matrix:** */
$\mathbf{P}^0_{\mathbf{k}} \leftarrow \text{diag}(f_{n,\mathbf{k}})$ (occupation numbers on diagonal)
$\mathbf{H}_{-1,\mathbf{k}} \leftarrow \mathbf{H}_{0,\mathbf{k}}$ (extrapolation seed)

/* **Time loop:** */
for $j = 1$ to N_{steps} **do**
  $t \leftarrow t_0 + j\cdot\Delta t$
  Extrapolate: $\mathbf{H}_{1,\mathbf{k}} \leftarrow 2\mathbf{H}_{0,\mathbf{k}} - \mathbf{H}_{-1,\mathbf{k}}$ [extrapolate_Hmatrix()]

/* **Self-consistent Hamiltonian loop:** */
  while $\|\mathbf{H}^{\text{new}}_1 - \mathbf{H}^{\text{old}}_1\|_\infty > H_{\text{thrs}}$ **do**
    Magnus operator: $\mathbf{\Omega}_{\mathbf{k}} \leftarrow -\imath\cdot\frac{1}{2}(\mathbf{H}_{0,\mathbf{k}} + \mathbf{H}_{1,\mathbf{k}})\Delta t$ [magnus()]
    Propagate density matrix: $\mathbf{P}^1_{\mathbf{k}} \leftarrow \mathbf{P}^0_{\mathbf{k}} + [\mathbf{\Omega}_{\mathbf{k}}, \mathbf{P}^0_{\mathbf{k}}] + \ldots$ [eldyn_ort() / commutp()]
    Update density: $\rho^\sigma(\mathbf{r}) \leftarrow \rho^\sigma_{\text{gs}}(\mathbf{r}) + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_n P^1_{nn,\mathbf{k}} |\psi_{n,\mathbf{k}}(\mathbf{r})|^2$ [GetNewRho_rmgtddft()]
    Update potentials: $V_{\text{H}}[\rho]$, $V^{\sigma}_{\text{xc}}[\rho^\uparrow, \rho^\downarrow]$
    Update Hamiltonian: $\mathbf{H}^{\text{new}}_{1,\mathbf{k}} \leftarrow \langle\psi_a|\hat{H}[\rho]|\psi_b\rangle$ [HmatrixUpdate()]
  end while

  Extract current: $J_\alpha(t) \leftarrow \text{Re}\sum_{\mathbf{k}} w_{\mathbf{k}} \,\text{Tr}[\mathbf{P}^1_{\mathbf{k}}\cdot\mathbf{J}^{\alpha}_{\mathbf{k}}]$
  (Optional) **Berry phase polarization:** $P_{\text{BP},\alpha}(t) \leftarrow \text{Tr}[\mathbf{P}^1_{\mathbf{k}}\cdot\mathbf{X}^{\alpha}_{\mathbf{k}}]$
  Advance: $\mathbf{P}^0 \leftarrow \mathbf{P}^1$; $\mathbf{H}_{-1} \leftarrow \mathbf{H}_0$; $\mathbf{H}_0 \leftarrow \mathbf{H}_1$
end for

/* **Postprocessing:** */
Fourier transform $J_\alpha(t)$ for each Cartesian direction α
Compute $\sigma_{\alpha\beta}(\omega)$, $\varepsilon_{\alpha\beta}(\omega)$

the free oscillations of the current density $\mathbf{J}(t)$ following the kick. The linear response regime is valid as long as the kick amplitude is small enough that the system does not leave the linear regime, which is straightforwardly verified by checking that the response is proportional to $|\mathbf{A}_0|$.

Diamagnetic current term. The full current density contains both paramagnetic and diamagnetic contributions (Section 2.5). In the present implementation, the diamagnetic term $\mathbf{J}_{\text{dia}} \propto \mathbf{A}(t)|\psi|^2$ is identically zero for all $t > 0$ because $\mathbf{A}(t > 0) = 0$ in the perturbative kick scheme. This is consistent with the linear response regime: the diamagnetic contribution to the linear response function is implicitly accounted for by the sum rule, and the paramagnetic current alone gives the correct optical conductivity.[?, ?] Simulations with a continuously applied driving field (e.g., a CW laser or a Gaussian pulse) would require retaining $\mathbf{A}(t)$ throughout the propagation and are beyond the scope of the present work.

First-order Magnus expansion. As in the finite-system implementation,[?] the Magnus operator is truncated to first order:

$$\mathbf{\Omega} \approx \mathbf{\Omega}_1 = -\imath\cdot\frac{1}{2}(\mathbf{H}_0 + \mathbf{H}_1)\Delta t. \qquad (10)$$

The second-order correction $\mathbf{\Omega}_2 = \frac{1}{12}\Delta t^2[\mathbf{H}_0, \mathbf{H}_1]$ is not included. The accuracy and stability of this scheme for the time steps used here was validated in paper 1 through energy conservation tests, and we demonstrate analogous stability for the periodic case in Section 4.

Norm-conserving pseudopotentials only. The current implementation supports only norm-conserving pseudopotentials (NCPP).[?, ?] Ultrasoft pseudopotentials (USPP)[24] introduce a non-trivial overlap matrix $\mathbf{S} \neq \mathbf{I}$ into the density matrix propagation (the nonorthogonal basis path), which requires the generalized propagation scheme described in Appendix A of paper 1. This extension is planned for future work.

BCH propagator for complex Hamiltonians. For k-points away from the Γ point, the Hamiltonian matrix $\mathbf{H}_{\mathbf{k}}$ is complex Hermitian. In this case, the BCH commutator expansion propagator (Ieldyn=1) is used throughout; the diagonalization-based propagator (Ieldyn=2) is not currently supported for complex matrices. The BCH expansion is unconditionally stable and has been shown to give results equivalent to exact diagonalization within the convergence threshold,[?] and we find no practical limitation from this choice.

3.3 Software Optimization and Parallelization

[Content outline – adapt from paper 1 Section 3.2 with] additions for the periodic case: - Same MPI + OpenMP + GPU parallelization as paper 1 - k-point parallelization: each MPI group handles a subset of k-points (pct.kpsub_comm); current is summed over k-points with weights (MPI_Allreduce over kpsub_comm, lines 828-829 in RmgTddft.cpp) - Spin parallelization: separate MPI communicator groups per spin channel (pct.spin_comm); spin channels communicate only for XC update (MPI_Allreduce over spin_comm, line 767) - The density matrix P and Hamiltonian H are distributed using ScaLAPACK layout (Sp object); size is numst x numst where numst = N_occ + N_virt - For VECTOR_POT mode: momentum matrix J_alpha is complex even at Gamma point (stored as complex<double> in Pxmatrix_cpu etc.) - The BCH commutator $[\Omega, P]$ dominates $O(N^3)$; same optimization as paper 1 (real/imaginary decomposition, BLAS ZGEMM/DGEMM) - Timing data for benchmark systems: [add when available]

[Note: include a brief timing comparison table analogous to Table 1 in paper 1, but for one or more periodic benchmark systems (e.g., Si supercells of increasing size) to demonstrate scaling.]

4 Numerical Results

We benchmark the periodic and spin-polarized RT-TDDFT implementation in `RMG` against the `Octopus` code[16] for four representative systems of increasing dimensionality and physical complexity: a one-dimensional hydrogen chain (H_2), a two-dimensional hexagonal boron nitride (hBN) monolayer, a three-dimensional silicon crystal, and the spin-polarized magnetic insulator CrI_3. These systems are chosen to isolate and validate specific aspects of the implementation: k-point sampling, anisotropic optical response, three-dimensional periodic boundaries, and spin-collinear propagation, respectively. The section concludes with a demonstration of the unique large-scale capability of `RMG` through calculations on nitrogen-vacancy (NV) centers in diamond at varying supercell sizes.

In all calculations, a momentum kick of amplitude E_0 = [value] a.u. was applied at $t = 0$ along each Cartesian direction independently, and the optical conductivity was obtained from the Fourier transform of the resulting current density. [Add: common computational parameters – time step, total propagation time, broadening gamma, XC functional, pseudopotential library used for all systems.]

4.1 One-Dimensional Periodic System: Hydrogen Chain

As the simplest possible periodic test system, we consider an infinite chain of hydrogen molecules aligned along the z axis with alternating bond lengths, modeled by a one-dimensional supercell with periodic boundary conditions. This system provides a stringent test of k-point sampling and the velocity gauge implementation in the limit of a quasi-one-dimensional geometry, where the longitudinal (z-direction) and transverse (x, y-direction) optical responses differ substantially.

[Computational details: - Supercell geometry: H–H bond length, H_2–H_2 separation - K-point sampling: [N_k points along z, Gamma in x and y] - Grid spacing: [value] a_0 - XC functional: [PBE or LDA] - Pseudopotential: [type] - Time step Δt: [value] a.u. - Total simulation time: [value] a.u. - Broadening γ: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both longitudinal and transverse components - K-point convergence: how many k-points needed for converged spectrum? - Physical interpretation of the spectral features - If Berry phase comparison is available: show agreement between current-integration and BP polarization as internal consistency check (Fruit 1 from CONTEXT.md) - If ground-state current convergence is shown: include as panel or note in text (Fruit 2 from CONTEXT.md)]

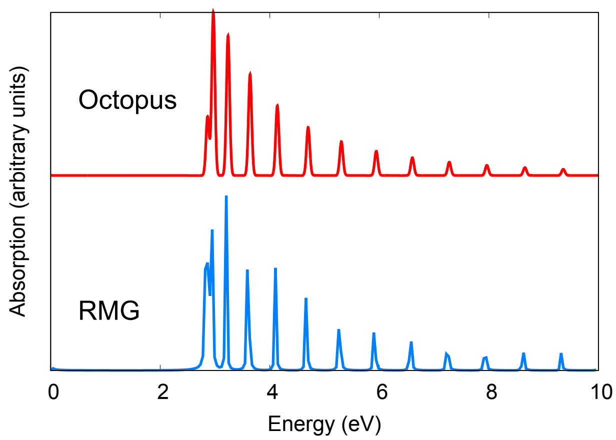

Figure 1: [Caption for H_2 chain optical conductivity figure. Should describe: (a) longitudinal $\sigma_{zz}(\omega)$ and transverse $\sigma_{xx}(\omega)$ components from RMG compared with Octopus; (b) convergence with k-point density if shown; labels for RMG and Octopus curves. Note anisotropy of 1D system as key feature to highlight.]

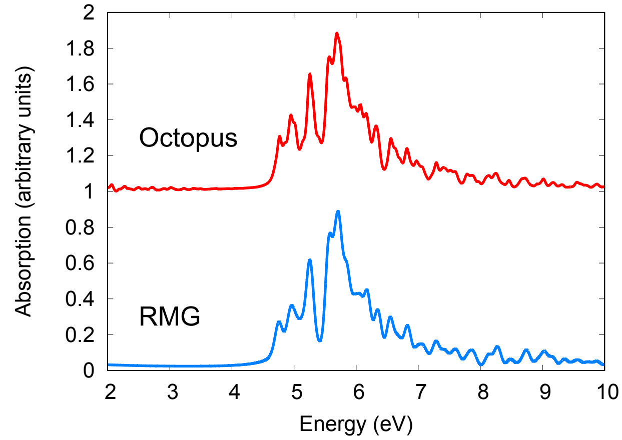

Figure 2: [Caption for hBN optical conductivity figure. Should describe: in-plane $\sigma_{xx}(\omega)$ from RMG and Octopus; out-of-plane component if shown; indicate the onset of absorption near the band gap. Highlight that $\sigma_{xx} = \sigma_{yy}$ (symmetry correctly reproduced) and $\sigma_{zz} \approx 0$.]

4.2 Two-Dimensional Periodic System: Hexagonal Boron Nitride

Hexagonal boron nitride (hBN) is a wide-gap insulating two-dimensional material with D_{6h} symmetry. Its optical response is highly anisotropic: the in-plane conductivity $\sigma_{xx}(\omega) = \sigma_{yy}(\omega)$ differs markedly from the out-of-plane component $\sigma_{zz}(\omega)$, the latter being negligible for photon energies below the interlayer excitations. This system tests the implementation of two-dimensional periodic boundary conditions and the symmetry enforcement of the conductivity tensor in `RMG`.

[Computational details: - Monolayer geometry: lattice constant [a] Å, vacuum layer [d] Å to prevent inter-layer interactions - K-point sampling: [N_k × N_k × 1] Monkhorst-Pack - Grid spacing: [value] a_0 - XC functional: [PBE] - Pseudopotential: [type] - Time step and total simulation time: [values]]

[Discussion points: - Agreement between RMG and Octopus for in-plane conductivity - Symmetry: show that $\sigma_{xx} = \sigma_{yy}$ (Fruit 3 from CONTEXT.md) - Physical features: onset at band gap, first excitonic peak if visible - Comparison with available experimental or LR-TDDFT data if relevant]

4.3 Three-Dimensional Periodic System: Silicon

Silicon in the diamond cubic structure ($Fd\bar{3}m$, space group 227) is the canonical benchmark for periodic electronic structure calculations. Its indirect band gap and well-characterized optical spectrum provide a rigorous test of the full three-dimensional periodic RT-TDDFT implementation. The cubic symmetry requires $\sigma_{xx}(\omega) = \sigma_{yy}(\omega) = \sigma_{zz}(\omega)$, providing an immediate symmetry validation.

[Computational details: - Unit cell: conventional 8-atom cell, lattice constant [a] Å- K-point sampling: [N_k × N_k × N_k] Monkhorst-Pack - Grid spacing: [value] a_0 - XC functional: [PBE] - Pseudopotential: [type, NCPP] - Number of occupied + virtual states: [N_occ + N_virt] - Time step Δt: [value] a.u. - Total simulation time: [value] a.u. ([value] fs) - Broadening γ: [value] eV]

[Discussion points: - Agreement between RMG and Octopus: peak positions, lineshapes - Symmetry: $\sigma_{xx} = \sigma_{yy} = \sigma_{zz}$ (Fruit 3) - Energy conservation test for periodic case (Fruit 5 from CONTEXT.md): run longer simulation and show no drift, analogous to paper 1 Fig. 3c - Berry phase comparison if included (Fruit 1) - Physical features: onset near direct gap, main optical peak structure]

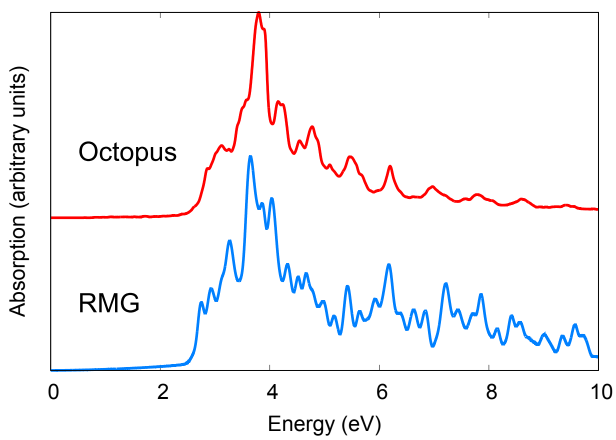

Figure 3: [Caption for Si optical conductivity figure. Should describe: $\sigma_{xx}(\omega)$ ($= \sigma_{yy} = \sigma_{zz}$ by cubic symmetry) from RMG compared with Octopus; indicate band gap region; optionally compare with experiment. If energy conservation panel included: describe it.]

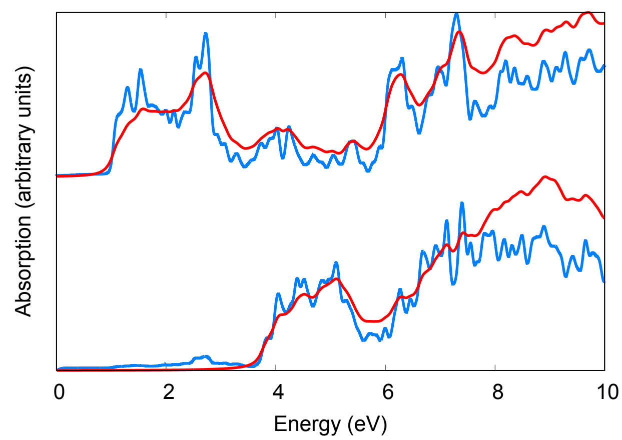

Figure 4: [Caption for CrI_3 optical conductivity figure. Should describe: spin-up and spin-down optical conductivities from RMG compared with Octopus; indicate the relevant spectral features (charge-transfer peaks, d-d transitions); note that spin channels are labeled consistently.]

4.4 Spin-Polarized System: CrI_3

Chromium triiodide (CrI_3) is a layered van der Waals magnetic insulator with a honeycomb lattice of Cr^{3+} ions, each carrying a local moment of approximately $3\,\mu_B$. In its ferromagnetic ground state, the spin-up and spin-down electronic structures differ, leading to distinct optical spectra for the two spin channels. This system provides a direct test of the spin-collinear RT-TDDFT extension: the two spin channels should propagate independently but with a common Hartree potential, yielding spin-resolved optical conductivities that can be compared with `Octopus` and with available experimental data.

[Computational details: - Unit cell: [describe geometry and magnetic ordering] - K-point sampling: [N_k × N_k × N_k] - XC functional: [PBE or PBE+U; specify U if used] - Pseudopotential: [type, NCPP] - Initial spin configuration: ferromagnetic, moments on Cr sites - Time step and total simulation time: [values] - Broadening: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both spin channels - Physical interpretation: which spectral features are spin-up vs spin-down? Charge-transfer excitations, Cr d-d transitions - Spin current $\mathbf{J}_{\rm spin} = \mathbf{J}^{\uparrow} - \mathbf{J}^{\downarrow}$ if computed (Fruit 4 from CONTEXT.md) - Comparison with available experimental optical data if relevant - Note: ECD/MCD spectra not computed in this work (magnetic perturbation formalism reserved for future publication)]

4.5 Large-Scale Demonstration: NV Center in Diamond

As a demonstration of the unique large-scale capability of `RMG`, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond. The NV center—a substitutional nitrogen atom adjacent to a carbon vacancy—is one of the most studied point defects in solid-state physics, of primary importance for quantum sensing and quantum information applications due to its long spin coherence time at room temperature.[?]

In a standard periodic DFT or RT-TDDFT calculation, the NV center is represented by a single defect in a periodically repeated supercell. The physical properties of the *isolated* defect are recovered only when the supercell is large enough that the interaction between the defect and its periodic images is negligible. Identifying this convergence threshold is both a practical necessity and a physically meaningful result: it establishes the minimum defect concentration at which the isolated-defect limit is reached.

We investigate this convergence by performing RT-TDDFT calculations on diamond supercells containing a single NV center, systematically increasing the cell size from [N_{min}] to [N_{max}] atoms. [Specific supercell sizes to be determined. Suggested sequence (subject to computational feasibility on Frontier): 64, 216, 512, 1000, [larger if possible] These correspond to defect concentrations of approximately [values] cm^{-3}.]

[Computational details: - Host: diamond cubic, lattice constant [a] Å- Defect: single NV^{-} center (negatively charged) - Spin state: triplet ground state ($S = 1$), ferromagnetic alignment - K-point sampling: Gamma point only for large supercells ([note if Gamma+additional k-points used for small cells]) - XC functional: [PBE] - Pseudopotential: [NCPP type] - Number of states: [N_occ + N_virt per spin] - Time step and total simulation time: [values] - Computational platform: Frontier at ORNL - Nodes used: [values per system size]]

[Figure placeholder: NV center convergence figure. Suggested panels: (a) Optical conductivity $\sigma(\omega)$ for each supercell size, showing convergence with increasing cell size (b) Convergence metric vs supercell size (e.g., peak position or integrated spectral weight) showing saturation (c) Optionally: timing data showing scaling with system size Label defect-related spectral features (zero-phonon line region, broad phonon sideband).]

Figure 5: [Caption for NV center convergence figure. Describe the system, the supercell sizes shown, the key spectral features, and the convergence behavior. Identify the minimum supercell size for which the isolated-defect optical spectrum is recovered within [tolerance].]

[Discussion points: - At what supercell size does the spectrum converge? What is the corresponding defect concentration threshold? - What is the physical origin of the convergence: direct defect-defect orbital overlap, or longer-range electrostatic interaction? - How does the spin-up and spin-down optical response differ? Does the convergence behavior differ between spin channels? - Timing data: how does the computational cost scale with cell size? At what size does the TDDFT step dominate over the SCF step? (Compare with paper 1 Table 1 format for Ag nanorods.) - This demonstration establishes the unique position of RMG for studying realistic defect concentrations in extended systems.]

5 Summary and Outlook

[To be written after all other sections are complete. Should summarize the key contributions, state the validated capabilities, and outline planned extensions: (1) continuous driving field / nonlinear response, (2) non-collinear spin RT-TDDFT, (3) Ehrenfest dynamics combined with spin, (4) ultrasoft pseudopotentials.]

6 Acknowledgements

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A Gauge Transformation of the Nonlocal Pseudopotential Term

[Full derivation of the gauge transformation of V^nl from length to velocity gauge, following the structure of Mattiat & Luber (2022) Appendix B but adapted to the Kleinman-Bylander form used in RMG. This is the extended algebra supporting section 2 subsection on current operator and nonlocal pseudopotentials.]

B Equivalence of Current-Based and Berry Phase Polarization

[Proof that the velocity gauge current J(t), when integrated, gives the same polarization as the King-Smith/Vanderbilt Berry phase formula. This supports the comparison presented in section **??**.]

C [Additional appendix if needed]

[Placeholder. May be used for: spin-collinear extension of the BCH scheme; extended orthogonalization proofs if needed beyond paper 1 appendix; or can be removed if not needed.]

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